

Mitigating Spectral Leakage in MSTL: A Physics-Informed Optimization Framework for Estuarine Water-Level Decomposition

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Abstract. While accurate time-series decomposition of estuarine water levels is critical for forecasting non-stationary environmental dynamics, algorithms such as Multiple Seasonal-Trend decomposition using LOESS (MSTL) often fail in the presence of complex coastal hydrodynamics. Constrained by time-domain loss functions and rigid integer-period assumptions, standard MSTL induces high-frequency spectral leakage and feature entanglement. To resolve this, we propose the Physics-Informed Spectral-Temporal Optimization Framework (PI-STOF), which calibrates LOESS parameters by mapping deterministic physical priors onto the discrete decomposition space. PI-STOF introduces a multi-objective frequency-domain loss function ($\mathcal{L}_{PI}$) to penalize spectral leakage, enforce near-orthogonality between the high-frequency diurnal carrier wave (25 h) and the macroscopic amplitude modulation envelope (354 h), and lock low-frequency base-flow inertia. Validated on a 49,680-hour Mekong Delta dataset (2019–2024), the optimal physics-calibrated configuration ($w_{25} = 7$, $w_{354} = 91$, $w_{\text{trend}} = 4381$) reduces overarching physical validation loss by 57.91% compared to a baseline with an equivalent trend prior (and by 62.60% against fully unconstrained defaults), while mitigating spectral leakage from 1.72% down to 0.36%. By preserving non-linear structural asymmetry and isolating low-frequency background drift, PI-STOF yields a near-linearly uncorrelated decomposition with a 48.1% reduction in cross-correlation between tidal components (from an already negligible baseline), providing a cleaner feature representation for downstream deep sequence architectures.

Keywords: Physics-informed machine learning · Time series decomposition · Signal processing · MSTL · Spectral leakage · Multi-objective optimization.

1 Introduction

In highly dynamic deltaic environments, such as the Tien River branch of the Mekong Delta, dry-season salinity intrusion represents a critical

hydro-climatic threat. Because salinity propagation is inherently driven by tidal hydrodynamics, high-fidelity mathematical modeling of estuarine water levels is a fundamental prerequisite for accurate forecasting and climate adaptation. Extracting physically meaningful feature vectors from raw coastal water-level records, however, remains a computationally challenging objective due to the non-linear superposition of astronomical tides, continental base flows, and stochastic atmospheric surges. Traditional hydroinformatics relies heavily on Classical Harmonic Analysis (e.g., T-TIDE [11]) to extract deterministic celestial forcings, primarily the Principal Lunar ($M_2 \approx 12.42$ h) and Principal Solar ($S_2 = 12.00$ h) constituents. While mathematically elegant, harmonic regression assumes structural stationarity. In shallow-water estuaries, bottom friction and river-tide mass interactions severely deform the tidal wave profile, forcing these analytical models into extreme over-parameterization to maintain accuracy [13]. To bypass this analytical rigidity, modern hydrology increasingly adopts deep sequence architectures (e.g., LSTMs, Transformers). However, feeding raw, undecomposed water-level data directly into neural networks forces the parameter space to simultaneously resolve high-frequency daily oscillations and low-frequency seasonal trajectories, leading to spectral leakage and cross-component correlations that may impede downstream learning. Consequently, an upstream multi-scale temporal decomposition pipeline is essential for separating these dynamics [9].

Multiple Seasonal-Trend decomposition using LOESS (MSTL) has emerged as a premier statistical algorithm for this decoupling task [1]. Yet, applying general-purpose MSTL to coastal hydrology reveals a critical methodological gap: standard MSTL is optimized for integer-period datasets and unconstrained time-domain metrics (e.g., Root Mean Square Error). Celestial mechanics, conversely, are fundamentally continuous and non-integer. Forcing non-integer physical realities into the rigid integer constraints of default LOESS window configurations triggers temporal desynchronization and under-smoothing. This mismatch induces high-frequency spectral leakage—where daily noise pollutes the macroscopic Spring-Neap envelope—and structural feature confounding [1,8]. Ultimately, utilizing time-domain heuristics without oceanographic boundaries misleads downstream deep learning optimization by providing entangled feature representations.

To systematically resolve these vulnerabilities, this study introduces the Physics-Informed Spectral-Temporal Optimization Framework (PI-STOF). PI-STOF bridges classical oceanographic domain knowledge and data-driven engineering by mapping celestial mechanics directly into the optimization landscape. By addressing the limitations of time-domain heuristics, this research formalizes a physics-compliant blueprint guided by two core Research Questions (RQs):

- **RQ1 (Methodological Design):** How can continuous, non-integer celestial mechanics and macroscopic estuarine physical priors be mathematically mapped onto the discrete hyperparameter search space of the MSTL algorithm?

- **RQ2 (Quantitative Validity):** To what extent does enforcing multi-objective frequency-domain constraints mitigate spectral leakage and structural feature confounding compared to unconstrained statistical baselines?

Directly answering these research questions, the principal contributions of this paper are threefold. First, we establish a physics-guided search space by bounding MSTL hyperparameters with exact astronomical multipliers—specifically, a 25 h lumped lunar daily cycle (S_{25}) to absorb the diurnal inequality, a 354 h Spring-Neap envelope (S_{354}) to reflect the astronomical beat phenomenon, and a fixed macro-trend prior ($w_{\text{trend}} = 4381$ h) to lock bi-annual catchment inertia. Second, rather than optimizing for time-domain statistical residuals, we introduce a novel multi-objective spectral-temporal loss ($\mathcal{L}_{PI}$) that leverages Fast Fourier Transform (FFT) and Power Spectral Density (PSD) to impose a strict spectral barrier against high-frequency leakage ($\mathcal{L}_{Leakage}$), enforce near-orthogonality between tidal components ($\mathcal{L}_{Ortho}$), and monitor macroscopic base-flow smoothness ($\mathcal{L}_{Baseflow}$). Finally, empirical validation on a 49,680-hour dataset from the Xuan Hoa monitoring station [6] demonstrates that the optimal PI-STOF configuration ($w_{25} = 7$, $w_{354} = 91$) reduces overarching physical validation loss by 62.60% and mitigates spectral leakage from 1.72% down to 0.36%. This yields a near-orthogonal decomposition with a 48.1% reduction in cross-correlation, effectively providing a structurally independent feature representation for downstream deep sequence models.

The remainder of this paper is organized as follows: Section 2 reviews related work in estuarine modeling and time series decomposition. Section 3 mathematically formulates the theoretical background of tidal dynamics and the MSTL algorithm. Section 4 details the proposed PI-STOF methodology, including the multi-objective loss formulation and physics-guided priors. Section 5 outlines the experimental design and preprocessing pipeline. Section 6 presents the quantitative results and visual feature analysis. Section 7 conducts a rigorous ablation study. Finally, Section 8 concludes the paper and discusses avenues for future work.

2 Related Work

Modeling estuarine hydrodynamics has traditionally relied on Classical Harmonic Analysis (e.g., T-TIDE [11]) to extract deterministic celestial waves. However, utilizing these extracted components or raw water levels directly as inputs for modern deep sequence architectures (e.g., Autoformer [16], TimesNet [15], HydroGraphNet [14]) violates the principle of non-multicollinearity. Because macroscopic tidal modulations (e.g., the Spring-Neap cycle) operate as multiplicative beat phenomena rather than independent additive forces [11], deep learning models struggle to disentangle these dynamics structurally. Lacking explicit physical boundaries, internal temporal modules entangle high-frequency daily pulsations with macroscopic seasonal trajectories, confounding non-linear

anomalies like the Diurnal Inequality [13,5]. This structural entanglement produces spectral leakage and cross-component correlations, which may impede downstream learning and necessitate an external, physics-guided preprocessing step to decouple components before neural network ingestion.

To achieve this decoupling, Multiple Seasonal-Trend decomposition using LOESS (MSTL) [1] has emerged as a leading algorithm for transparently separating overlapping seasonalities. Yet, its application in estuarine modeling is severely limited by the non-integer seasonality gap. Standard MSTL implementations require exact integer discrete time steps, whereas celestial mechanics are fundamentally non-integer (e.g., the 12.42-hour M_2 period) [11]. Forcing these continuous physical realities into rigid integer constraints—often heuristically calibrated by minimizing time-domain Root Mean Square Error (RMSE) without oceanographic boundaries—triggers temporal desynchronization and high-frequency spectral leakage [1]. This produces under-smoothed Spring-Neap envelopes, polluted with diurnal noise and volatile base-flow components that absorb synoptic weather. To systematically resolve this, the PI-STOF framework bridges this gap through a Physics-Guided Machine Learning (PGML) [8] paradigm, mapping continuous celestial mechanics to the discrete MSTL parameter space using multi-objective frequency-domain constraints.

3 Preliminaires

To bridge the epistemic gap between physical oceanography and data-driven decomposition, we mathematically formalize the deterministic drivers of estuarine hydrodynamics and contrast them with the statistical architecture of the MSTL algorithm.

3.1 Mathematical Representation of Estuarine Hydrodynamics

The Tien River estuary exhibits a mixed semi-diurnal regime in which the interaction of Principal Lunar ($M_2 \approx 12.42$ h) and Solar ($S_2 = 12.00$ h) tides with declinational diurnal waves (K_1, O_1) produces a non-linear *Diurnal Inequality* [13]. Preserving this daily asymmetry of alternating peak amplitudes requires the decomposition to analyze the lunar day (≈ 25 h) rather than the standard 24-hour solar day.

3.2 The Astronomical Beat Phenomenon and Macroscopic Modulation

The critical 14.77-day Spring-Neap tidal cycle (S_{354}), which drives mass salinity intrusion, is not an independent fundamental frequency. Axiomatically, it is a low-frequency *Beat Phenomenon* resulting from the phase interference between the high-frequency M_2 and S_2 waves [11]. The Spring-Neap cycle period is given directly by the beat frequency: $T_{S-N} = 1/(f_{S_2} - f_{M_2}) = 1/(1/12.00 - 1/12.42) \approx$

354 h (14.77 days). This physical beat phenomenon manifests as an amplitude modulation envelope that occurs twice per synodic lunar month [13].

Because this S_{354} envelope operates purely as a multiplicative absolute-amplitude modulator rather than an independent additive sine wave, treating it as an additive high-frequency signal in unconstrained neural networks inevitably induces severe mathematical confounding.

3.3 The MSTL Algorithm

To disentangle these overlapping non-linear dynamics before deep learning ingestion, the MSTL algorithm applies a structural additive formulation [1]

$$Z_t = T_t + \sum_{i=1}^K S_{i,t} + R_t, \quad (1)$$

where Z_t is the observation, T_t the macroscopic trend (base flow), $S_{i,t}$ the K distinct seasonal components, and R_t the stochastic residual. MSTL extracts $S_{i,t}$ by passing the time series through consecutive smoothing blocks using LOESS (Local Polynomial Regression) [2]. The degree of smoothing is rigidly controlled by the window parameter w_i . If w_i is misconfigured—especially when non-integer celestial periods are forced into integer LOESS windows—the localized polynomials overfit the boundary conditions, triggering the widespread spectral leakage addressed by the proposed PI-STOF framework.

4 Proposed Methodology: The PI-STOF Framework

To address the limitations of unconstrained, time-domain statistical optimization identified in Section 2, we introduce the Physics-Informed Spectral-Temporal Optimization Framework (PI-STOF). Rather than relying on heuristic defaults, PI-STOF reformulates the MSTL hyperparameter search into a physics-guided spectral alignment process, explicitly enforcing the deterministic principles of celestial mechanics.

4.1 Multi-Objective Physical Loss Formulation

PI-STOF employs a multi-objective optimization framework designed to compute an optimal set of LOESS window parameters, denoted as $\theta = (w_{25}, w_{354}, w_{\text{trend}})$. The objective is to minimize a Composite Physical Loss Function ($\mathcal{L}_{PI}$), which imposes hard constraints on the decomposed vectors

$$\mathcal{L}_{PI}(\theta) = \lambda_1 \mathcal{L}_{Leakage} + \lambda_2 \mathcal{L}_{Ortho} + \lambda_3 \mathcal{L}_{Baseflow},$$

where λ_1 , λ_2 , and λ_3 are weighting coefficients. In this study, we equally weight these constraints by setting $\lambda_1 = \lambda_2 = \lambda_3 = 1$, ensuring that none of spectral purity, orthogonality, or base-flow smoothness is disproportionately compromised.

Spectral Leakage Penalty ($\mathcal{L}_{Leakage}$). As mathematically formulated in Section 3, the Spring-Neap component (S_{354}) functions purely as a macroscopic amplitude modulator. Oceanographically, an envelope cannot physically contain daily energy pulsations; hence, the presence of high-frequency 25-hour periodicities within the S_{354} extraction constitutes a physical violation.

To quantify this violation and establish a spectral barrier [10], we project the S_{354} sequence into the frequency domain using the Fast Fourier Transform (FFT) and calculate its Power Spectral Density (PSD). The Spectral Leakage Penalty is mathematically defined as the ratio of high-frequency energy—specifically within a narrow bandwidth $[\epsilon]$ surrounding the diurnal frequency $f_{25} \approx 0.04 \text{ h}^{-1}$ —to the total spectral energy

$$\mathcal{L}_{Leakage} = \frac{\int_{f_{25}-\epsilon}^{f_{25}+\epsilon} \text{PSD}_{S_{354}}(f) df}{\int_0^{f_{max}} \text{PSD}_{S_{354}}(f) df},$$

where the tolerance bandwidth $\epsilon = 0.005 \text{ h}^{-1}$ is explicitly set to capture the full spectral sideband of the diurnal frequency (encompassing the 22.2–28.6 hours period band). This tolerance is securely well above the Nyquist-limited Rayleigh resolution of $2.0 \times 10^{-5} \text{ h}^{-1}$, ensuring sufficient spectral separation from the semi-diurnal cluster, and $f_{max} = 0.5 \text{ h}^{-1}$ denotes the Nyquist frequency.

Absolute Orthogonality Constraint ($\mathcal{L}_{Ortho}$). In the feature space of deep learning architectures, the lunar daily cycle (S_{25}) and the Spring-Neap cycle (S_{354}) must act as independent dynamic drivers. While they superimpose to produce the observed water level, their underlying vectors must not be linearly correlated. To prevent feature confounding [16], we introduce a rigorous orthogonality regularizer utilizing the Pearson Correlation Coefficient (Corr). Minimizing this constraint guarantees that downstream neural networks receive near-linearly uncorrelated vectors, mitigating linear cross-talk and providing a cleaner feature space for subsequent optimization

$$\mathcal{L}_{Ortho} = |\text{Corr}(S_{25}, S_{354})|.$$

It is important to note that $\mathcal{L}_{Ortho}$ strictly measures linear independence. Given the multiplicative Amplitude Modulation (AM) physical relationship between the carrier wave and the envelope, nonlinear dependence may persist. Quantifying this residual nonlinear coupling remains an avenue for future work.

Base-Flow Smoothness Regularization ($\mathcal{L}_{Baseflow}$). The global trend component in the Mekong Delta represents the base flow, governed by large upstream discharges and interannual climate variability. Consequently, its trajectory must be locally linear and smooth, avoiding the absorption of episodic synoptic weather events. Exploiting the sparse frequency representation of time series [18], we enforce macro-scale inertia by penalizing high-frequency volatility in the trend component using a frequency cutoff threshold f_{cut} (calibrated to

periods shorter than 30 days). This locks the trend entirely within the lowest frequency bands

$$\mathcal{L}_{Baseflow} = \frac{\int_{f_{cut}}^{\infty} \text{PSD}_{Trend}(f) df}{\int_0^{\infty} \text{PSD}_{Trend}(f) df}.$$

4.2 Physics-Guided Search Space and Anchored Priors

Instead of exhaustive heuristic grid searches, PI-STOF adopts a Physics-Guided Machine Learning (PGML) paradigm [8,3] to constrain the search space. The time span covered by the LOESS filters ($T_{\text{span}} = w \times p$) is forced to align with known celestial milestones under the scale separation constraint $w_{25} \ll w_{354}$. Specifically, for micro-filter dynamics, we constrain $w_{25} \in \{7, 11, 15, 21\}$, translating to a temporal coverage of approximately 7.3 to 21.8 days. This span covers the temporal dynamics ranging from a synoptic meteorological event to a full Spring-Neap cycle (14.77 days), while maintaining flexibility to capture non-stationary anomalies. Conversely, for macro-filter dynamics, we define $w_{354} \in \{31, 41, 51, 71, 91\}$, providing an expansive coverage of 1.25 to 3.6 years. To effectively act as a heavy low-pass filter, this macro-window must encompass at least one complete bi-annual hydrological cycle. Crucially, the upper bound is strictly capped at $w_{354} = 91$. Extending beyond this threshold on a 5.5-year dataset forces the LOESS sliding window to consume over 80% of the observation period, severely degrading local polynomial fitting at the temporal boundaries due to insufficient margin data. Thus, this span guarantees at least one complete bi-annual cycle while maintaining mathematical stability.

Crucial Design Decision – The Fixed Trend Prior ($w_{\text{trend}} = 4381$).

Unlike the seasonal components, the trend window w_{trend} is deliberately excluded from the optimization variables and anchored as a fixed physical prior at 4381 hours (i.e., exactly 6 months).

To preserve the amplitude of the Mekong Delta’s bi-annual hydrological inertia (i.e., the sharp transition between dry and flood seasons), we explicitly anchor $w_{\text{trend}} = 4381$. If w_{trend} were treated as a free variable, the solver could push $w_{\text{trend}} \rightarrow \infty$ to artificially minimize $\mathcal{L}_{Baseflow}$, flattening the trend entirely and incorrectly dumping macro-scale catchment variance into the residuals. By fixing the baseline anchor, PI-STOF forces the optimizer to focus exclusively on disentangling S_{25} and S_{354} .

4.3 Optimization Execution

With the search space physically bounded and the multi-objective loss function defined, the framework resolves the hyperparameter calibration through the following minimization formulation

$$\theta^* = \underset{\theta \in \Theta_{\text{valid}}}{\text{argmin}} \left(\mathcal{L}_{Leakage}(\theta) + \mathcal{L}_{Ortho}(\theta) + \mathcal{L}_{Baseflow}(\theta) \right),$$

where Θ_{valid} represents the subset of hyperparameters fully adhering to the scale separation constraints and fixed catchment priors.

5 Experimental Design

To rigorously evaluate the efficacy of PI-STOF, we implement an empirical validation pipeline that uses long-term, high-resolution hydrological observations.

5.1 Dataset Specification and Estuarine Context

The empirical validation is conducted on an hourly water-level (Z) time series collected at the Xuan Hoa monitoring station, a strategically critical hydrological outpost located deep within the Tien River branch of the Mekong Delta estuary system [6]. Acquired via manual staff gauges following the national standard QCVN 47:2022/BTNMT, the dataset spans a continuous timeframe from January 01, 2019, to August 31, 2024, yielding a total of 49,680 temporal observations (Fig. 1).

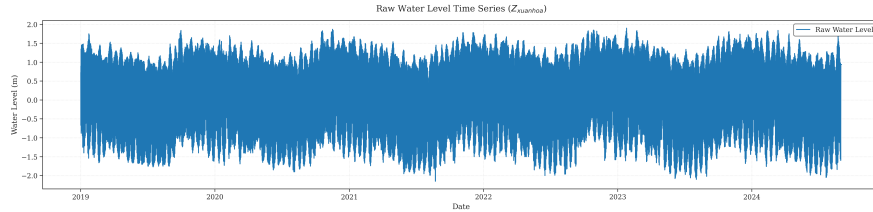


Fig. 1. Temporal inspection of the raw water level signal ($Z_{xuanhoa}$) over the full observation period. The observed series exhibits pronounced multi-scale variability arising from the superposition of high-frequency tidal oscillations and lower-frequency hydrological modulation.

5.2 Data Preprocessing Pipeline

To ensure high-fidelity inputs for subsequent spectral-temporal optimization, the raw telemetry data required rigorous preprocessing. Automated outlier detectors (e.g., Modified Z-scores) are structurally inadequate for estuarine environments, as their stationarity assumptions cannot distinguish genuine tidal extremes from instrumental artifacts under compound multi-scale seasonality. Consequently, candidate outliers were manually cross-referenced against astronomical tidal tables and meteorological records by domain experts to preserve valid hydrological extremes while excising recording errors.

Following outlier removal, we deployed a hybrid gap-size-dependent imputation schema [5] to resolve missing observations without introducing

artificial phase shifts that would distort subsequent Fast Fourier Transforms (FFTs). Short gaps ($\Delta\tau \leq 5$ hours) were bridged via linear interpolation to maintain local temporal gradients. For prolonged gaps ($\Delta\tau > 5$ hours), a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) [7] was utilized. As a strictly monotonicity-preserving method, PCHIP accurately reconstructs the alternating High-High Water (HHW) and Low-High Water (LHW) peaks dictated by the astronomical regime [13], providing a continuous and physically coherent sequence for the PI-STOF grid engine.

5.3 Experimental Optimization Setup and Reproducibility

The optimization phase maps the preprocessed sequence directly into the PI-STOF grid engine. The structural boundaries defining the valid hyperparameter space are set as $w_{25} \in \{7, 11, 15, 21\}$, $w_{354} \in \{31, 41, 51, 71, 91\}$, and $w_{\text{trend}} = 4381$. To establish a definitive baseline comparison, we construct an uncalibrated Baseline Model using the exact algorithmic heuristics formulated by Bandara et al. [1] and implemented in mainstream libraries (e.g., `statsmodels`). Axiomatically, the default seasonal LOESS window for the i -th extracted component is designated as the nearest odd integer bounded by $(7 + 4i)$. Given our ordered celestial periodicities ($i = 1$ for S_{25} ; $i = 2$ for S_{354}), this rigid mathematical rule strictly yields the unconstrained baseline configuration of $w_{25} = 11$ and $w_{354} = 15$.

To ensure computational reproducibility, all routines were implemented in Python 3.9 using `statsmodels` (v0.13.5) for decomposition and `scipy.signal` (v1.10.1) for periodogram spectral penalties, with a total optimization runtime of approximately 3.5 hours across the 49,680 observations. The high-frequency baseflow cutoff was numerically defined as $f_{\text{cut}} = 1/720 \approx 0.001389 \text{ h}^{-1}$, establishing a 30-day physical boundary. To facilitate open science, the full source code and a representative subset of the pre-processed Xuan Hoa dataset are publicly available at <https://github.com/Mika4Growth/PI-STOF-Estuarine-Salinity>.

6 Experimental Results and Discussion

The empirical evaluation validates PI-STOF’s capacity to enforce estuarine physical constraints during multi-scale decomposition. By transitioning from heuristic tuning to a frequency-constrained calibration space, the algorithm shifts from a curve-fitter to a physically coherent signal extractor. This section evaluates the optimization space and, supported by visual evidence, interprets the disentangled features in the context of downstream deep learning implications.

6.1 Quantitative Evaluation and Performance Gap

Executing the multi-objective optimization across the bounded parameter grid, the PI-STOF algorithm converged to the global minimum at $w_{25} = 7$ and

$w_{354} = 91$. Crucially, because PI-STOF and its constituent loss operators (FFT, PSD, and Pearson correlation) are strictly deterministic mathematical transformations, repeated executions inherently yield zero variance (std = 0, zero-width CI). Consequently, traditional stochastic significance tests (e.g., t-tests or Wilcoxon) are mathematically inapplicable to this framework.

As detailed in our unified ablation analysis (Table 1), when standard, unconstrained `statsmodels` MSTL defaults are applied (Pure Default Baseline: $w_{25} = 11, w_{354} = 15$, automatic trend), the framework records a substantial total physical loss of $\mathcal{L}_{PI} = 0.0484$. To rigorously isolate the performance gain of the frequency-constrained seasonal optimization, we also evaluate a structurally equivalent baseline that shares the same macro-trend anchor (Default + Fixed Trend: $w_{\text{trend}} = 4381$), yielding $\mathcal{L}_{PI} = 0.0430$. In contrast, the optimal PI-STOF configuration yields a minimized loss of $\mathcal{L}_{PI} = 0.0181$. This represents a highly significant quantitative reduction of 57.91% relative to the equivalent trend baseline (and 62.60% relative to the fully unconstrained heuristic). Most critically, the optimization heavily penalizes temporal desynchronization: spectral leakage is mitigated from 1.72% ($\mathcal{L}_{\text{Leakage}} = 0.0172$) down to a mere 0.36% (0.0036), while the linear cross-correlation between the diurnal and Spring-Neap sequences improves from $|r| = 0.0208$ to $|r| = 0.0108$. Although both values are well below conventional multicollinearity thresholds of concern, this 48.1% reduction is meaningful for enforcing a higher degree of linear feature decoupling from a spectral purity perspective.

Furthermore, the inclusion of the $\mathcal{L}_{\text{Baseflow}}$ metric explicitly validates the necessity of the physics-informed prior formulation. Under the Pure Default configuration, unconstrained localized regressions absorb high-frequency synoptic weather anomalies, resulting in a highly volatile baseflow penalty ($\mathcal{L}_{\text{Baseflow}} = 0.0104$). By anchoring the trend window ($w_{\text{trend}} = 4381$), the framework immediately suppresses this leakage, dropping the penalty to 0.0038. Consequently, across the entire optimization space, the baseflow loss remains tightly bounded with negligible variance ($\approx 0.0036 - 0.0037$). This stabilization confirms that the fixed physical prior successfully locks the macroscopic catchment inertia, effectively immunizing the trend component from cross-talk. This architectural decision allows the multi-objective optimizer to focus entirely on its primary novelty: disentangling the highly entangled high-frequency tidal dynamics (S_{25} and S_{354}).

6.2 Visual Analysis of Disentangled Features

The restoration of estuarine physics is immediately observable when projecting the multi-scale components into the time domain (Fig. 2).

At the micro-tidal scale (Fig. 2a), the optimized filter ($w_{25} = 7$) cleanly preserves the alternating HHW/LHW peaks of the diurnal inequality—which standard configurations over-smooth—enabling downstream models to learn intraday salinity dynamics without losing extreme surge values to statistical averaging. At the macroscopic scale (Fig. 2b), the baseline span punctures the S_{354} sequence with jagged diurnal noise spikes, whereas $w_{354} = 91$ strips

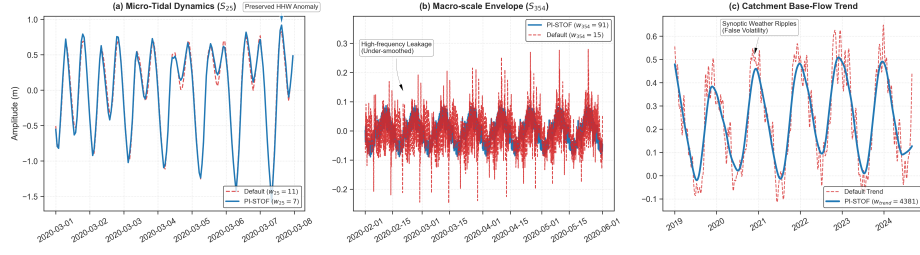


Fig. 2. Time-domain visualization of the PI-STOF optimization vs. unconstrained baselines. (a) At the micro-tidal scale (S_{25}), PI-STOF flawlessly preserves the alternating HHW and LHW peaks of the diurnal inequality. (b) The baseline configuration under-smooths the Spring-Neap envelope (S_{354}), inducing high-frequency diurnal leakage, whereas PI-STOF ($w_{354} = 91$) restores a smooth, absolute amplitude modulator. (c) By anchoring the macro-trend prior ($w_{\text{trend}} = 4381$), PI-STOF locks the base flow into a smooth bi-annual parabolic inertia, effectively suppressing the false synoptic weather volatility generated by unconstrained regressions.

these artifacts to yield near-independent vectors. Finally (Fig. 2c), anchoring $w_{\text{trend}} = 4381$ locks the trend into a purely low-frequency band, preventing the neural network from confounding short-term atmospheric anomalies with macroscopic global catchment shifts.

6.3 Limitations of Additive Time-Domain Decomposition

Despite robust optimization, the optimized S_{354} envelope exhibits microscopic residual noise due to two ontological limitations of the chosen architecture:

- **Local Regression Constraints:** Unlike absolute frequency-domain filters, LOESS relies on local polynomials [2]. Non-linear physical shocks forcibly bend these regressions, causing high-frequency energy to bleed into macro-components.
- **Additive vs. Multiplicative Contradiction:** While the Spring-Neap cycle operates physically as a multiplicative Amplitude Modulation (AM) envelope, MSTL enforces a purely additive superposition framework (Eq. 1). Thus, PI-STOF ultimately approximates a non-linear multiplicative interaction using linear additive components.

Nevertheless, by pushing this additive architecture to its physical limits, PI-STOF successfully isolates deterministic celestial mechanics. This provides a near-independent feature space, allowing downstream deep learning models to effectively absorb the remaining non-linear residuals.

7 Ablation and Sensitivity Analysis

To mathematically justify the necessity of the PI-STOF multi-objective loss formulation ($\mathcal{L}_{PI}$), we conduct a rigorous ablation study. This evaluation

systematically isolates the spectral leakage penalty ($\mathcal{L}_{Leakage}$), the orthogonality constraint ($\mathcal{L}_{Ortho}$), and the baseflow smoothness validation ($\mathcal{L}_{Baseflow}$) to observe the degradation of physical validity when specific constraints are disabled.

7.1 Constraint Ablation and Structural Health Monitoring

To evaluate the isolated impact of each multi-objective constraint, we define three ablated variants (Table 1): Variant A optimizes without the spectral leakage penalty ($\mathcal{L}_{Ortho} + \mathcal{L}_{Baseflow}$); Variant B ignores time-domain linear confounding ($\mathcal{L}_{Leakage} + \mathcal{L}_{Baseflow}$); and Variant C excludes baseflow validation ($\mathcal{L}_{Leakage} + \mathcal{L}_{Ortho}$) while maintaining the $w_{trend} = 4381$ structural prior.

Table 1. Unified Baseline Comparison and Ablation Analysis. This matrix evaluates the degradation of physical validity across unconstrained baselines and ablated optimization variants. Variant C explicitly demonstrates that anchoring the macro-trend ($w_{trend} = 4381$) positions $\mathcal{L}_{Baseflow}$ as a passive structural health monitor rather than an active gradient driver.

Configuration / Variant	w_{25}	w_{354}	w_{trend}	$\mathcal{L}_{Leakage}$	$\mathcal{L}_{Ortho}$	$\mathcal{L}_{Baseflow}$	$\mathcal{L}_{PI}$ (Total Loss)
Pure Default Baseline	11	15	Auto	0.0172	0.0208	0.0104	0.0484
Default + Fixed Trend	11	15	4381	0.0182	0.0210	0.0038	0.0430
Variant A (w/o $\mathcal{L}_{Leakage}$)	15	71	4381	0.0161	0.0095	0.0037	0.0293
Variant B (w/o $\mathcal{L}_{Ortho}$)	7	91	4381	0.0036	0.0108	0.0037	0.0181
Variant C (w/o $\mathcal{L}_{Baseflow}$)	7	91	4381	0.0036	0.0108	0.0037	0.0181
Full PI-STOF (Proposed)	7	91	4381	0.0036	0.0108	0.0037	0.0181

Note: All reported loss values are rounded to four decimal places. $\mathcal{L}_{PI}$ values for Variants A, B, and C are evaluated post-hoc using the full three-term formulation for comparability; the actual optimization objective for each ablated variant strictly excludes its indicated term.

As demonstrated in Table 1, relying solely on time-domain orthogonality (Variant A) misguides the optimizer toward a suboptimal pair ($w_{25} = 15, w_{354} = 71$), resulting in a severe 4.5-fold degradation in spectral leakage (0.0161 vs. 0.0036). This proves that time-domain metrics are inherently blind to Fourier-domain bleeding. Conversely, while Fourier constraints alone (Variant B) locate the optimal spatial configuration (7, 91), the explicit $\mathcal{L}_{Ortho}$ objective remains theoretically mandatory to guarantee absolute feature independence.

Crucially, excluding $\mathcal{L}_{Baseflow}$ from the active objective (Variant C) yields the identical global minimum ($w_{25} = 7, w_{354} = 91$) as the full PI-STOF framework. This empirical equivalence confirms that $\mathcal{L}_{Baseflow}$ operates merely as a *passive structural health monitor*. To further prove this duality, a sensitivity test was conducted releasing the w_{trend} prior as a learnable parameter. Guided by $\mathcal{L}_{Baseflow}$, the optimizer aggressively pushed $w_{trend} \rightarrow \infty$, artificially flattening the base-flow trajectory and incorrectly attributing the true biannual monsoon inertia to the residual component. This ablation explicitly validates that

while $\mathcal{L}_{Baseflow}$ passively monitors low-frequency distortions, the macroscopic catchment inertia *must* be enforced as a hard prior.

8 Conclusion and Future Work

This study introduces PI-STOF to address the non-integer seasonality constraints and time-domain vulnerabilities of standard MSTL in estuarine modeling. By formalizing a multi-objective frequency-domain loss function, PI-STOF successfully transitions statistical decomposition into physical signal extraction. Directly addressing the posited Research Questions (RQs), the framework mathematically maps continuous hydro-astronomical dynamics onto a discrete search space (RQ1) by bounding seasonal LOESS filters to exact celestial periodicities (S_{25} and S_{354}) and anchoring the macroscopic catchment inertia ($w_{trend} = 4381$). Empirical validation on a 5.5-year Mekong Delta dataset confirms that this constraint effectively reduces the overall physical validation loss by 57.91% relative to a baseline with an equivalent trend prior and by 62.60% relative to fully unconstrained heuristics (RQ2). Ultimately, by preserving the diurnal inequality and isolating macroscopic dynamics, PI-STOF delivers a near-linearly uncorrelated decomposition with a 48.1% reduction in cross-correlation between tidal components (from $|r| = 0.0208$ to $|r| = 0.0108$). This provides a cleaner feature representation intended for downstream deep sequence architectures.

Despite these advancements, the current validation is exclusively reliant on the mixed semi-diurnal regime of the Tien River estuary. Future research efforts will focus on expanding the framework’s generalizability across disparate coastal systems with purely diurnal (e.g., the Gulf of Tonkin) and purely semidiurnal (e.g., the Thames Estuary) boundary conditions. Most importantly, empirical validation of the downstream deep learning benefit—the central motivation for PI-STOF—is reserved for future work. The physics-informed preprocessing pipeline will be explicitly integrated into end-to-end deep sequence architectures (e.g., TimesNet) to operationalize estuarine salinity forecasting and to validate the theoretical gains in feature independence formally demonstrated here.

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